

MTD367 TMA01

Question 1(a): Arrays vs Dictionaries in Swift

- Show iteration
- Show safe access

Array	Dictionary
Ordered Sequence	Unordered key-value store
Duplicates allowed	Keys unique, values can duplicate
Access by integer index	By key (must be hashable)
$O(n)$ typical linear search	$O(1)$ average for key lookup
Shopping cart where order and duplicates matter	Inventory where fast access by ID or name matters

Question 1(b): Palindrome function

- Write a function
- Return bool
- Include reasonable amount of test cases

Question 2: Food delivery app data

- Use Decimal for price: Rounding, Display with 2 decimal point precision
- Explain orderID: String vs Int vs UUID
- items as Tuple or struct is ok
- Use date (Foundation framework) as deliveryDate
 - Can delivery date change? Must every order have a delivery date?
- Use enum for OrderStatus. Why?
- Update status regression?

Question 3(a): View, State, Navigation

- Real World Examples are sometimes missed
- View:
 - Declarative description of UI
 - Lightweight struct value
 - What to draw given a current state
 - Automatic updates and organization
 - View modifiers

- State
 - @State stores local, view-owned mutable data
 - @State ensures the value persists across re-renders
 - Marking properties with @State lets SwiftUI manage their lifetime and trigger body updates when they change

- ObservableObject
 - A shared, reference-type model used across multiple views
 - Properties marked with @Published automatically notify observers on change
 - @ObservedObject → The model is passed in from elsewhere
 - @StateObject → when the view creates/owns the model

- **NavigationStack**
 - A container that manages a navigation path (stack of views)
 - Works with a `NavigationPath` for programmatic navigation and type-safe destinations
 - Keeps track of the user's back/forward navigation automatically

- `NavigationLink`
 - A declarative way to create a tappable element that pushes a new view onto the `NavigationStack`
 - Can link by:
 - Destination view: `NavigationLink("Label", destination: DetailView())`
 - Value binding (with `NavigationPath`): `NavigationLink(value: item)`

Question 3(b) & 3(c)

- No need for Login checking
- No need for SwiftData or CoreData
- Explain flow with a diagram
- Long Press Gesture with Animation feedback
- Multiple Navigation Stacks